Deploying Machine Learning to Detect Social Phobia in UK School Students: A Pathway to Early Intervention and Educational Impact

Abstract

Social phobia, or social anxiety disorder, is a persistent challenge for school-aged students in the United Kingdom. It limits participation, undermines confidence, and can have long-term consequences for both academic achievement and social development. Current identification practices are inconsistent and often rely on subjective observation, leaving many students unnoticed until difficulties become severe. This project proposes the creation and validation of a machine-learning screening tool for deployment in schools and by local authorities. The tool will be designed to detect early indicators of social phobia in anonymised student data, offering educators and practitioners a practical means to recognise risk earlier. Validation will be achieved by employing the Social Phobia Inventory (SPIN) questionnaire administered within partner schools, ensuring the tool is tested against established measures of social anxiety and refined for accuracy, interpretability, and ethical integrity.

1. Introduction

Social phobia, also known as social anxiety disorder, is one of the most common anxiety conditions among young people. Estimates suggest that between 7–12% of individuals experience social anxiety at some point in their lives (Connor et al., 2000; Stein et al., 2000). A recent UK populationbased cohort study confirmed that anxiety disorders are widespread in middle childhood and early adolescence, with social anxiety contributing significantly to impairment in school and social functioning (MoralesMuñoz et al., 2022). For schoolaged children, social phobia manifests as an intense fear of social interactions, often resulting in avoidance behaviours, academic difficulties, and reduced participation in peer activities. These challenges can have longterm consequences for educational attainment, social development, and overall wellbeing (Schwartz et al., 2019). Despite its prevalence, identification of social phobia in schools remains inconsistent. Current practices often rely on subjective observation by teachers and/or parents, which can delay recognition until difficulties become severe. There is therefore a pressing need for more reliable, scalable, and ethically robust approaches to early detection.

This project aims to address that gap by developing and validating a machinelearning screening tool that can be deployed in schools and by local authorities, providing educators and practitioners with a practical means of recognising risk earlier and supporting timely intervention. Validation will be achieved using the Social Phobia Inventory (SPIN), a psychometrically validated selfreport measure of social anxiety symptoms (Connor et al., 2000). Guided by this context, the study will address two central research questions:

- How accurately can machinelearning models classify social phobia risk in schoolaged children using SPIN data compared to traditional observation methods?
- To what extent can early detection through a validated screening tool inform schoollevel interventions and contribute to safeguarding and educational policy?

2. Problem Statement

Social phobia among school-aged children in the UK is underidentified due to reliance on inconsistent, subjective observation, leaving many students without timely support.

3. Objectives

• To develop a machinelearning screening tool capable of detecting early indicators of social phobia in anonymised student data.

• To validate the tool using the Social Phobia Inventory (SPIN) questionnaire within partner schools to ensure accuracy and reliability.

• To deploy the tool in collaboration with schools and local authorities for practical use in early identification.

• To disseminate findings to academic, educational, and policy stakeholders to inform consistent screening protocols and practice.

4. Literature Review

a. Prevalence of Social Phobia Among School Students

Social phobia, or social anxiety disorder, is a common mental health concern among adolescents and schoolaged children. Research indicates that approximately 7–12% of children and adolescents experience social phobia, with its prevalence often peaking during early adolescence (Connor et al., 2000; Stein, Torgrud and Walker, 2000). Social phobia can significantly affect students' academic performance, social development, and emotional wellbeing, making it a critical issue for intervention within educational environments. Recent studies indicate that children with social anxiety may struggle to engage in class discussions, participate in group activities, and interact socially with peers, leading to isolation and lower academic achievement (Kashdan and Rottenberg, 2010). These behaviours can persist into adulthood, contributing to ongoing mental health difficulties such as depression and generalised anxiety disorder (Ginsburg et al., 2014). Despite its prevalence and longterm consequences, identification of social phobia in schools remains inconsistent, relying largely on subjective observation. This highlights a critical research gap: the absence of scalable, validated screening tools that can systematically detect early indicators of social phobia within educational settings. Addressing this gap through the development and validation of a machinelearning screening tool, grounded in the Social Phobia Inventory (SPIN), offers a novel pathway to improve early detection, intervention, and longterm outcomes.

b. Current Approaches for detecting Social Phobia

In the United Kingdom, detection of social phobia in children and adolescents primarily relies on structured diagnostic interviews and validated selfreport measures such as the SPIN (Connor et al., 2000) and broader anxiety scales like the Screen for Child Anxiety Related Emotional Disorders (SCARED) (Muris et al., 1999). These tools have demonstrated strong psychometric properties in clinical and research contexts, and NICE guidelines recommend their use for recognition and assessment of social anxiety disorder (NICE, 2013). However, their application in schools remains limited, as they are resourceintensive, require trained clinicians, and are not scalable for routine screening (Stein, Torgrud and Walker, 2000). Consequently, many cases go undetected until difficulties become severe, underscoring the need for a scalable, validated, and ethically robust screening tool that can be embedded within educational environments to support earlier recognition and intervention.

c. Machine Learning in Social Phobia Assessment and Intervention

Recent studies have explored the potential of machine learning (ML) to assess and predict social anxiety in schoolaged populations. Hwang and Lee (2019) applied clustering algorithms to responses from SPIN and the Social Anxiety Scale for Children (SASC), successfully identifying subgroups of students based on distinct anxiety patterns and demonstrating the feasibility of predictive analytics for early diagnosis. Similarly, Jabbar et al. (2018) used ML models to predict the risk of social phobia

in adolescents from selfreported data on social interactions and psychological traits, achieving an accuracy rate of 95%. These findings underscore the promise of ML in enhancing early identification and intervention. Yet, current ML applications remain experimental and constrained to research contexts, with several limitations. Most rely heavily on selfreported data, which can be biased and difficult to scale in school environments. Models often prioritise prediction accuracy without addressing interpretability, ethical oversight, or integration into safeguarding frameworks. Critically, there is still no validated, schoolembedded ML tool designed for systematic deployment in UK educational settings.

5. Research Methodology

The study will be conducted in three sequential stages: data collection, data analysis and model development, and profile categorisation with tool deployment, spanning a total of five months to ensure systematic progression from survey administration to the creation of a validated, webbased screening tool for social phobia in UK schools.

Data Collection (Month 1–1.5)

• Approach welcome and interest UK schools to collect anonymous survey data from students using the SPIN questionnaire.

• Gather demographic information (e.g., age, gender, academic performance) to examine correlations with social phobia.

• Ensure ethical approval, safeguarding protocols, and parental consent where required.

Data Analysis and Model Development (Month 2–3.5)

• Clean and prepare survey and demographic data, addressing missing values and ensuring data quality.

• Train supervised machine learning models to classify students by risk level of social phobia.

• Identify key predictors of social anxiety severity from SPIN responses and demographic variables..

• Evaluate model accuracy, sensitivity, and specificity using crossvalidation and consultation with psychologists and mental health experts.

Tool Deployment and Validation (Month 4–5)

• Define risk categories (e.g., low, moderate, high) to guide potential school interventions.

• Validate outputs with psychologists and mental health experts to ensure the tool reflects realworld patterns of social phobia

• Deploy the screening model as a secure, userfriendly web application accessible to schools, enabling systematic early detection and reporting.

6. Expected Outcomes and Project Significance

The project will deliver a validated, webbased screening tool for social phobia that UK schools can adopt to systematically identify atrisk students using SPIN data enhanced by machine learning. Beyond immediate schoollevel application, the tool provides a scalable model that can inform national safeguarding and educational policy, strengthening early intervention strategies and contributing to REF impact through demonstrable improvements in practice, adoption, and policy alignment.

7. FollowUp Tracking, Impact Measurement and Findings Dissemination

• FollowUp Tracking will involve monitoring tool usage and student outcomes in partner schools to evidence sustained impact.

• Impact Measurement will be assesses using adoption rates, accuracy of risk identification, and expert feedback to validate effectiveness.

• Findings will be shared through academic publications, policy briefs, and practitioner workshops to maximise reach and policy influence.

8. Data Privacy and Ethical Considerations

All data collected will be handled in strict compliance with UK GDPR and school safeguarding protocols, ensuring anonymity and confidentiality for all student participants. Ethical approval and parental consent will be secured, with clear procedures to protect student welfare and prevent any misuse of sensitive information.

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